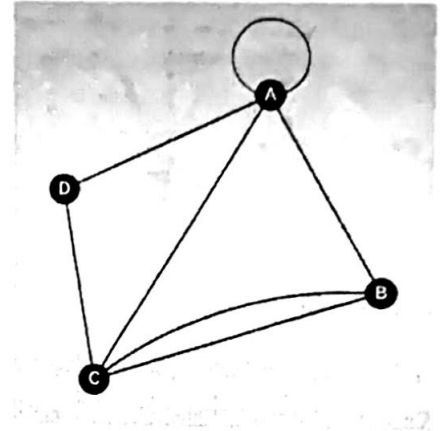


	Mathematics Applications Unit 3 2020	Year Weighting	Working Time	Total Time	YEAR 12
	Test 1	8%	55 min		
		Resources Allowed	Marks		
	Name: SOLUTIONS	No Calculator No notes	60	55	
	Full working must be shown for all questions (or parts) worth more than 2 marks				

Question One

(1, 2, 3, 2 = 8 marks)

The graph shown represents four buildings A, B, C and D with connecting walkways, at a local school.



(a) Why is the graph planar?

(b) Show that the graph satisfies Euler's formula.

(c) Construct the adjacency matrix for the graph.

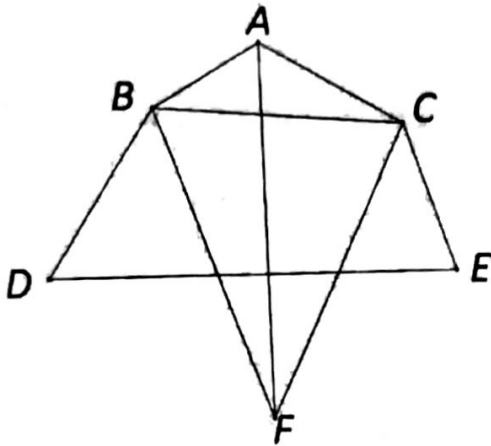
A student wishes to carry out closed walks of length two from Building A to Building B.

(d) List all his possible walks.

Question Two

(2 mark)

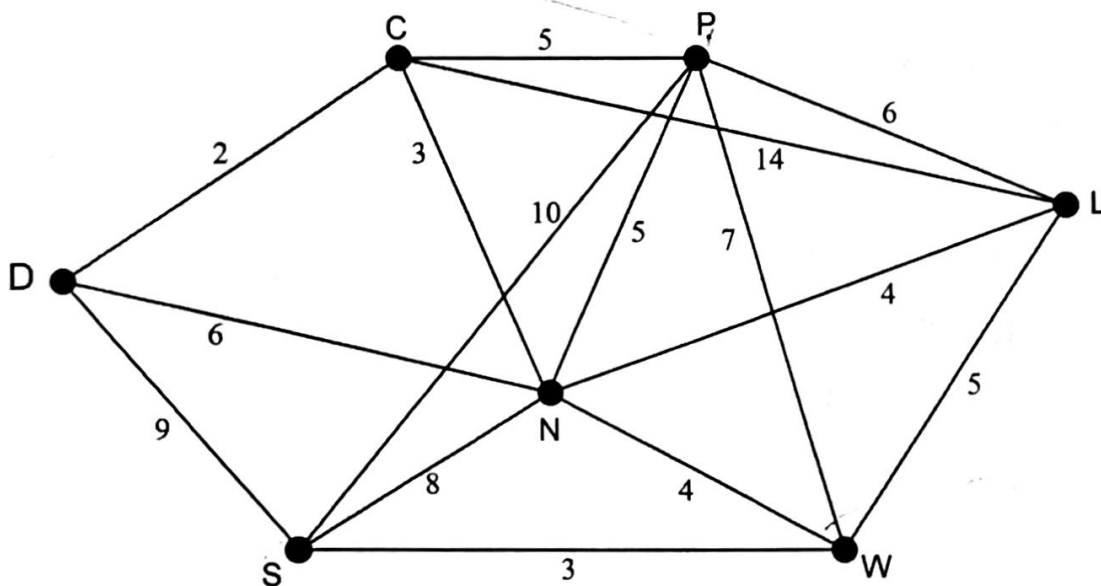
Redraw the following graph as a planar graph.



Question Three

(1, 1, 2, 2 = 6 marks)

Each morning Bert leaves the depot (D) to deliver milk to cafés in his area. He visits each café once and has produced a map showing the number of cafés, the roads linking them and the lengths (kilometres) of these roads. His map is represented by the graph below.



(a) (i) Identify the number of vertices on this graph.

(ii) Identify the number of edges on this graph.

Is the graph planar? Justify your answer.



(b) This graph is described as a simple and connected graph. Explain why this description can be applied.



Question Four

(2, 2, 1 = 5 marks)

In a commercial complex, buildings are named by their position (E.g. North 1 and East 2). North 1 connects with North 2, North 3, East 1, South 1 and West 1. East 1 connects with East 2. South 2 connects with South 1 and West 1. West 1 connects with West 2.

(a) Draw a graph of the complex.



(b) Which edges are bridges? Explain.

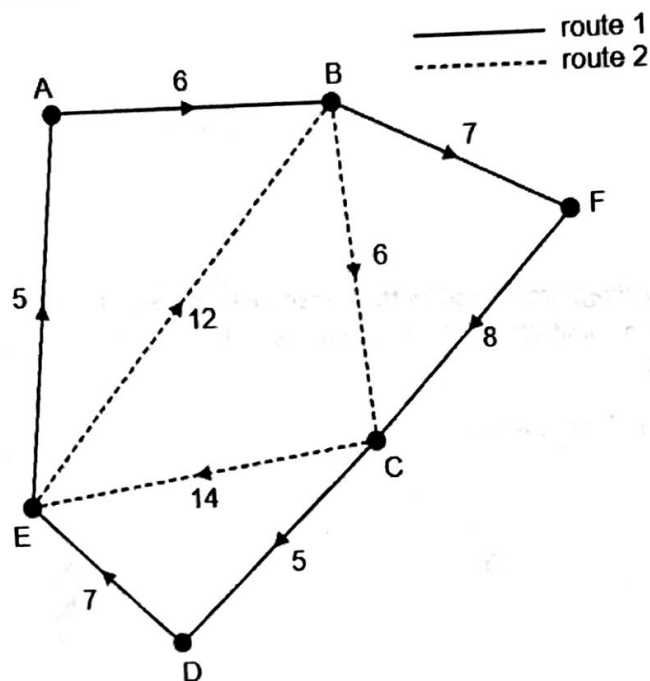


(c) Draw the graph K_4 .

(2, 2, 2, 2 = 8 marks)

Question Five

A bus company conducts 'jump on, jump off' sightseeing tours, during which tourists can get on and off buses at any of the designated attractions as many times as they like during the same day. The weighted digraph below shows Attractions A to F, along with the time (in minutes) that a bus takes to travel between the attractions. The bus company operates two different circuits around the city, each shown differently below.



(a) Complete the adjacency matrix for the digraph.

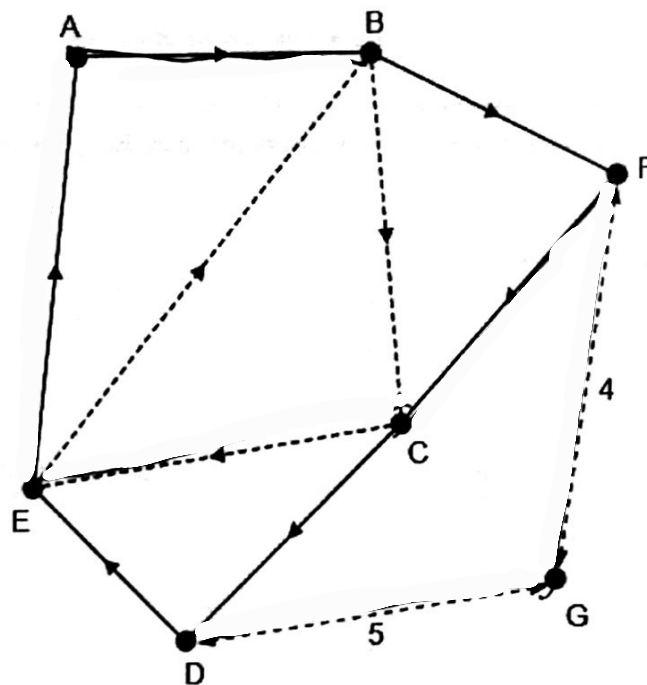
(a)

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	0	0	1	0	0	1
C	0	0	0	1	1	0
D	0	0	0	0	1	0
E	1	1	0	0	0	0
F						

, Mai gets on a bus at E. She travels directly to B, changes bus, and continues travelling BFCDEA. Explain why her route is a trail.

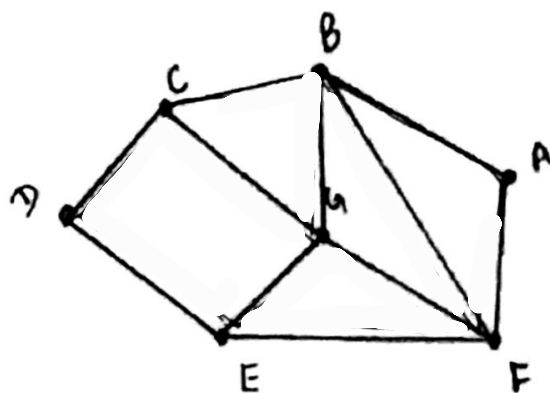
(c) Vinh has just visited Attraction C and wishes to visit Attraction B next. Determine the route he should take to arrive at B in the shortest travelling time. State the time taken.

The bus company plans to include another attraction, G. Instead of adding another circuit, they will provide shuttle buses between F and G (four minutes in each direction) and D and G (five minutes in each direction), as shown below.



(d) Toshi is at Attraction D. He wants to complete a Hamiltonian path. State the route he should take.

Question Six



(a) Determine the degree of each vertex in the network above and use the table to record your results.

Vertex	A	B	C	D	E	F	G
Degree							

(b) This network forms a semi-Eulerian trail

(i) What property of the vertices supports the claim that this network forms a semi-Eulerian trail?

(ii) Describe the semi-Eulerian trail for the above network by listing the vertices of the trail in the order visited.

Question Seven

(2 marks)

The subjects studied by four students are shown in the table. Represent this using a bipartite graph.

Student	Subjects
Amy	Maths, Chemistry, Media
Beth	Physics, Chemistry
Charles	Art, Media
Darren	Maths, Art

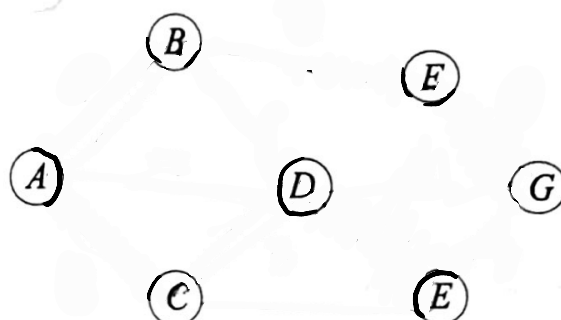
Question Eight

(3, 2, 2, 2 = 9 marks)

The table below shows the distances, in kilometres, along direct footpaths between the checkpoints A, B, C, D, E, F and G at a national park.

	B	C	D	E	F	G
A	4	4	7	-	-	-
B		-	1	-	3	-
C			3	5	-	-
D				2	-	5
E					-	2
F						3

(a) Complete the graph below to show this information.



A park worker is at Checkpoint D and needs to visit all the other checkpoints just once and then return to D.

(b) The worker leaves Checkpoint D, walks to Checkpoint A and then visits the rest of the checkpoints just once before returning to D. List the other checkpoints visited, in order, and state the total length of this route.

(c) Is the trail in (b) a Hamiltonian path or cycle? Explain your answer.

(d) A shorter trail, starting and finishing at D, is possible for the worker. List the order the checkpoints should be visited to achieve the smallest total length and state this length.

Question Nine

(2, 2, 1, 2, 2, 2, 1, 1 = 11 marks)

(a) A planar graph has five faces and five vertices, A, B, C, D and E.

(i) Determine the number of edges for this graph.

ii) Draw the planar graph in the space below.

(iii) Determine a Hamiltonian cycle for the graph, giving your answer as a sequence of vertices.

(iv) Is the graph Eulerian, semi-Eulerian or neither? Justify your answer.

(b) (i) A simple connected graph contains five vertices. Determine the minimum and the maximum number of edges it contains.

(ii) A simple connected graph contains n vertices. Determine the minimum number of edges it contains.

(iii) What name is given to the simple connected graph with the maximum number of edges possible